

Industrial Internship Report on " Automatic File Organizer using Python"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Automatic File Organizer using Python". It automatically organizes files into folders based on their file extensions using Python. The application reduces manual effort, improves file management, and keeps directories clean and organized by categorizing documents, images, videos, audio files, archives, and other file types automatically.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

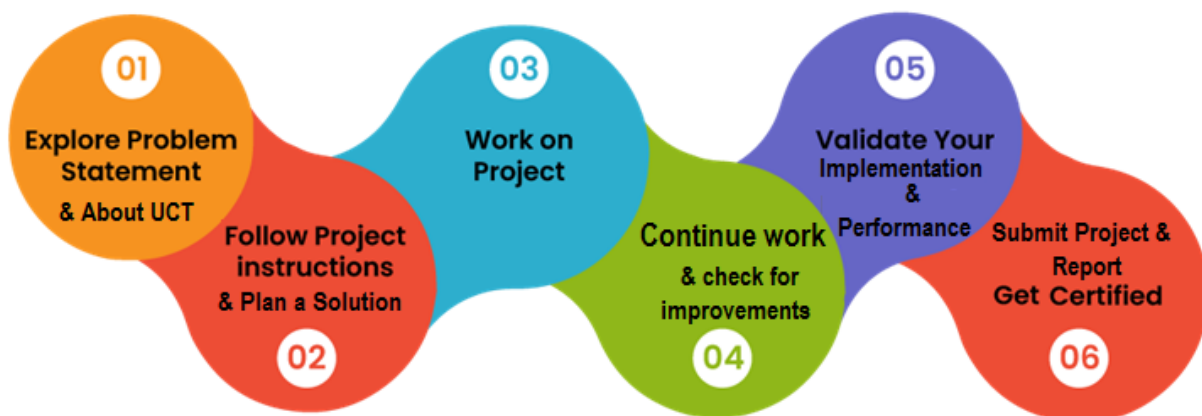
This internship provided me with an excellent opportunity to gain practical knowledge and understand how software development projects are planned, designed, implemented, tested, and documented in an industry-oriented environment. Throughout the six-week internship conducted by **upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT)**, I was able to apply my Python programming skills to solve a real-world problem.

The internship played an important role in my career development by helping me understand industry standards, project management, debugging techniques, version control using Git and GitHub, software testing, and technical documentation. It also improved my problem-solving abilities and strengthened my confidence in developing practical software solutions.

The project assigned during this internship was "**Automatic File Organizer using Python.**" The objective of the project was to develop a Python application that automatically organizes files into separate folders based on their file extensions. The application categorizes documents, images, videos, audio files, archives, and other file types, thereby reducing manual effort and improving file management.

This internship opportunity provided by **upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT)** allowed me to work on an industry-relevant project while following a structured software development approach. The internship enhanced both my technical and professional skills.

The overall program was planned in multiple stages. Initially, I understood the project requirements and problem statement. After planning the solution, I implemented the application using Python modules such as **os**, **shutil**, and **pathlib**. The project was then tested, improved through debugging and code optimization, documented properly, uploaded to GitHub, and finally prepared for submission.



Throughout this internship, I learned Python file handling, modular programming, software testing, GitHub version control, debugging, documentation, and professional coding practices. These learnings will be beneficial for my future academic projects, internships, and software engineering career.

I would like to express my sincere gratitude to **upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT)**, my mentors, trainers, and faculty members for their continuous guidance and support throughout the internship.

Finally, I encourage my juniors and fellow students to actively participate in internships and practical projects, as they provide valuable real-world experience and significantly improve technical, professional, and problem-solving skills.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



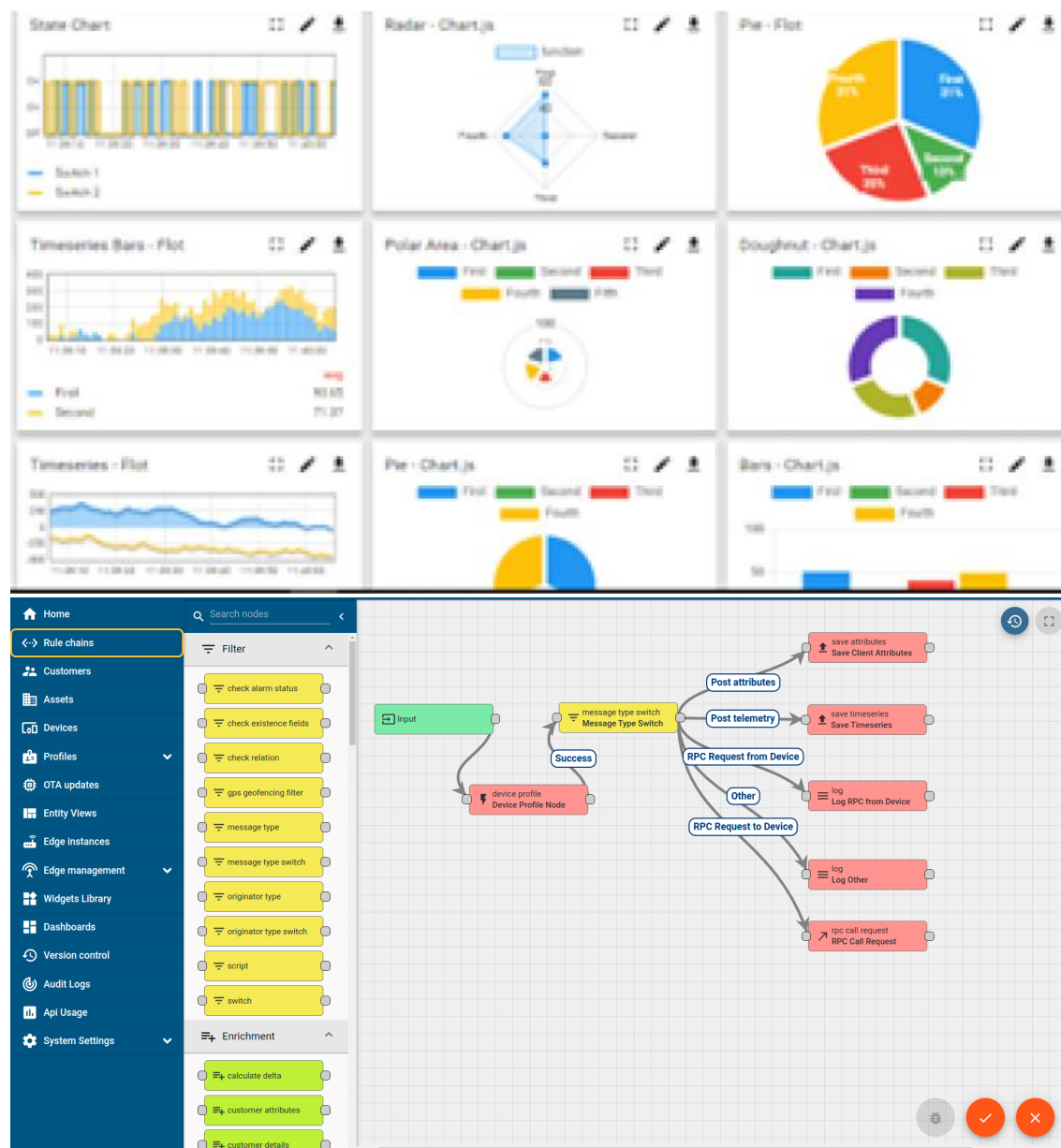
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



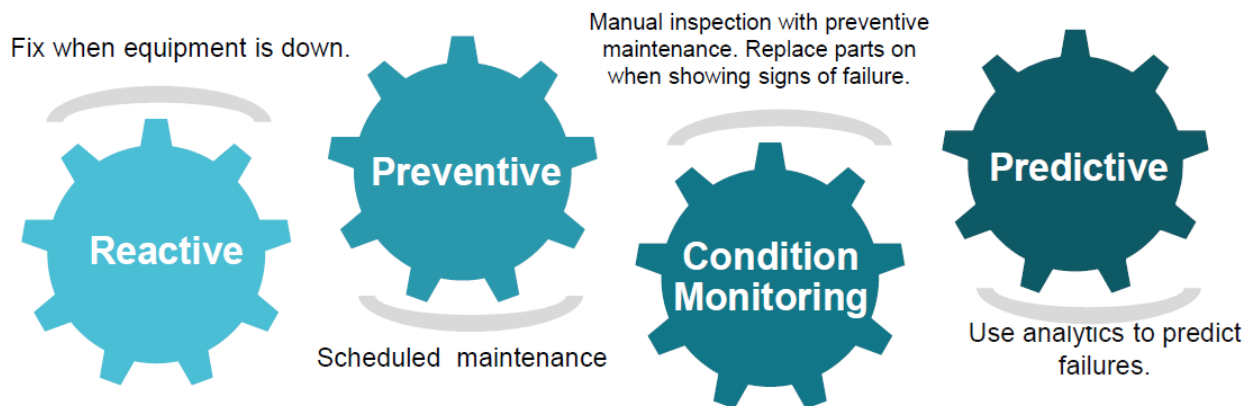


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

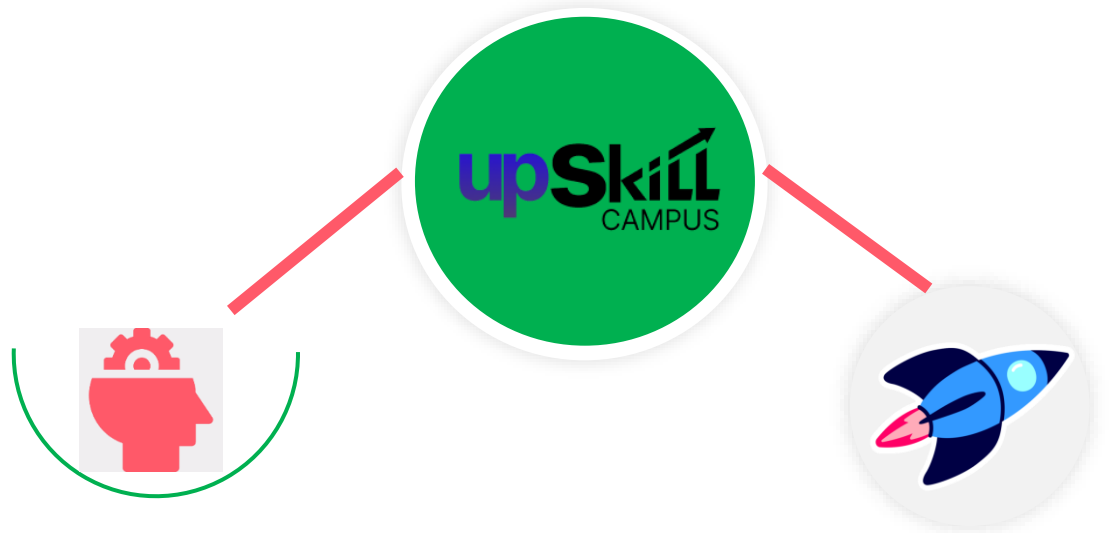
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

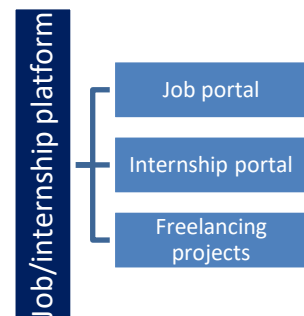
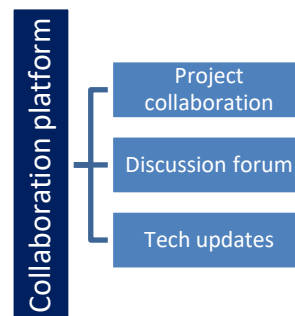
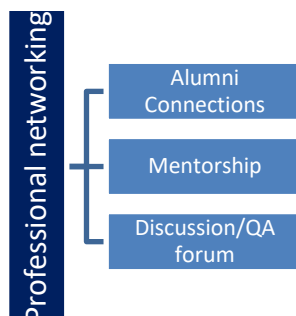
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation. *Python 3 Documentation*. <https://docs.python.org/3/>
- [2] Python shutil Module Documentation. <https://docs.python.org/3/library/shutil.html>
- [3] Python os Module Documentation. <https://docs.python.org/3/library/os.html>

2.6 Glossary

Terms	Acronym
Operating System	OS
Graphical User Interface	GUI
Python	PY
Application Programming Interface	API
Integrated Development Environment	IDE

3 Problem Statement

Description: The file organizer is a Python project that helps users organize their files in a directory. It scans a specified directory, categorizes files based on their type (e.g., images, documents, videos) and moves them into respective folders.

Scope: The scope of this project involves designing a user interface to specify the directory to organize implementing functions to identify file types and create folders and developing a file-moving algorithm to organize files into the appropriate folders.

4 Existing and Proposed solution

Existing Solutions

Several existing methods are available for organizing files. Users commonly organize files manually using Windows File Explorer by creating folders and moving files individually. Some users also rely on third-party file management applications that provide file sorting and organization features. Cloud storage platforms offer basic file organization capabilities, but they often require manual folder management. Although these methods are useful, they require continuous user involvement and become inefficient when handling a large number of files.

Limitations of Existing Solutions

The existing solutions have several limitations. Manual organization is time-consuming and requires significant effort. There is a high possibility of human error while sorting files, especially when dealing with large directories. Many third-party applications are either paid or include unnecessary features for simple file organization tasks. Existing methods also lack flexibility and require frequent manual maintenance to keep folders organized.

Proposed Solution

The proposed solution is an **Automatic File Organizer using Python**. The application automatically scans a selected directory, identifies files based on their extensions, creates category folders if they do not already exist, and moves each file into the appropriate folder. The project is developed using Python's standard libraries such as **os**, **shutil** and **pathlib**, making it lightweight, efficient, and easy to maintain. The application organizes documents, images, videos, audio files, archives, and other file types automatically, reducing manual effort and improving file management.

Value Addition

The proposed solution offers several advantages over existing methods. It automates file organization, reduces manual work, saves time, and improves productivity. The application is lightweight, free to use, easy to customize, and built entirely using Python's standard libraries without requiring additional software. Its modular design also makes it easy to extend with new file categories and additional features in the future.

4.1 Code submission: GitHub Repository <https://github.com/vinayak-30-debug/upskillCampus/blob/main/main.py>

4.2 Report : [https://github.com/vinayak-30-debug/upskillCampus/blob/main/AutomaticFileOrganizer Vinayak USC UCT.pdf](https://github.com/vinayak-30-debug/upskillCampus/blob/main/AutomaticFileOrganizer%20Vinayak%20USC%20UCT.pdf)

5 Proposed Design/ Model

5.1 Design Flow

The proposed solution follows a simple and efficient workflow to automatically organize files into categorized folders based on their file extensions. The system begins by accepting the directory selected by the user. It scans all files present in the directory and identifies the extension of each file. Based on predefined categories stored in the configuration file, the application checks whether a suitable destination folder already exists. If the folder does not exist, it is created automatically. The file is then moved to its corresponding folder. This process continues until all files in the selected directory have been organized. Finally, the application displays a completion message indicating that the organization process has finished successfully.

5.2 High Level Diagram

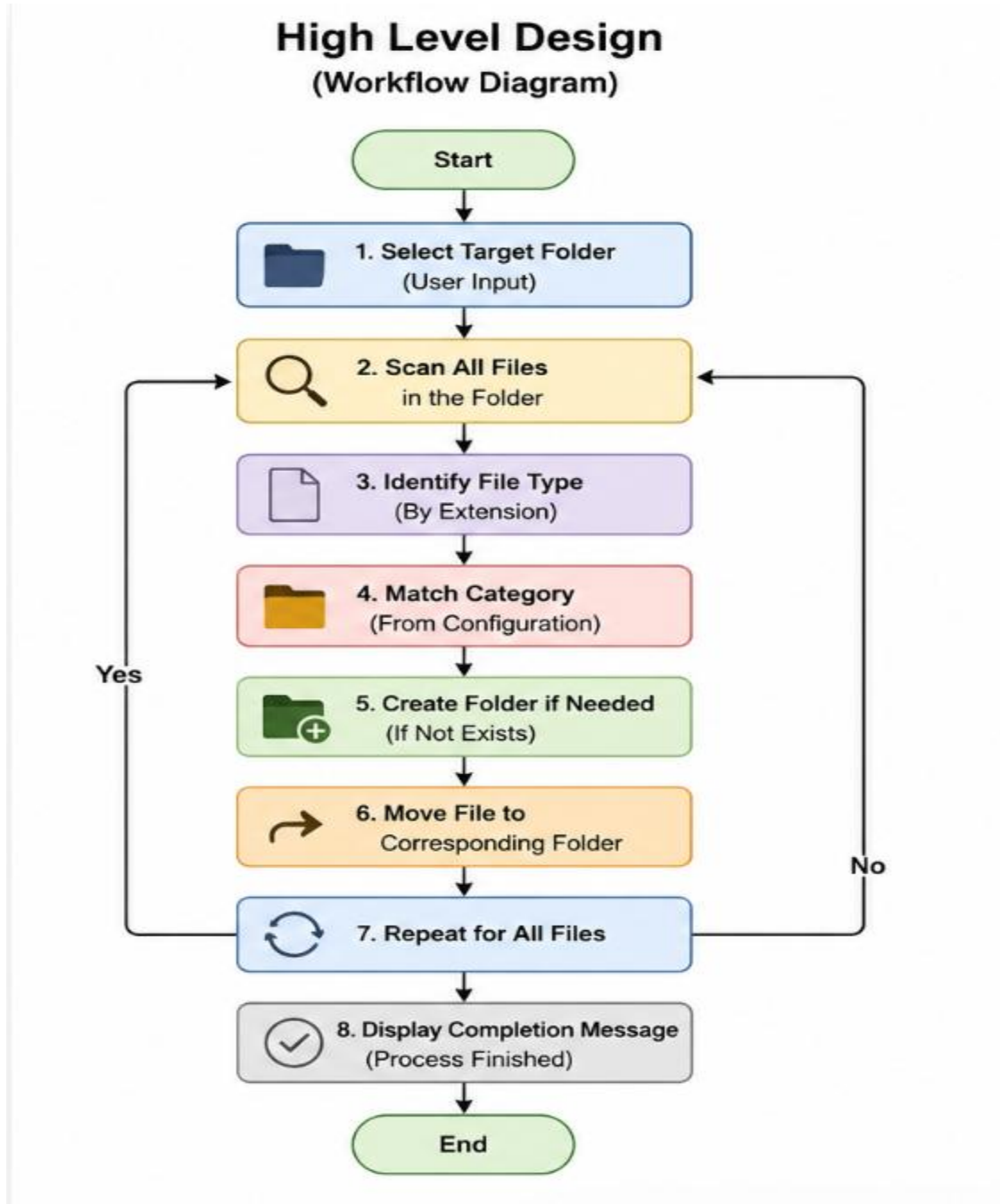


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Low Level Diagram

Low Level Design (Module Diagram)

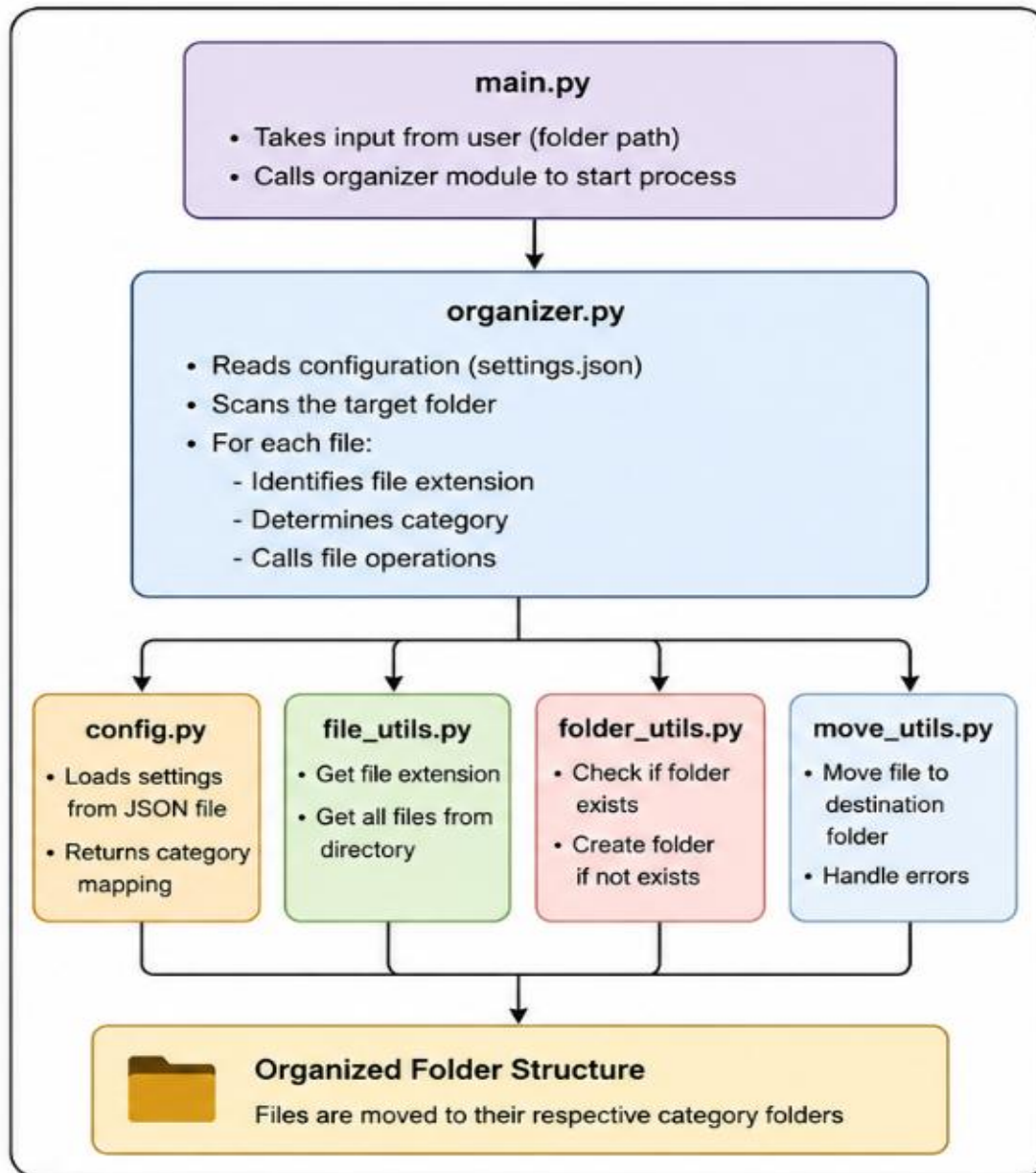


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.4 Interfaces

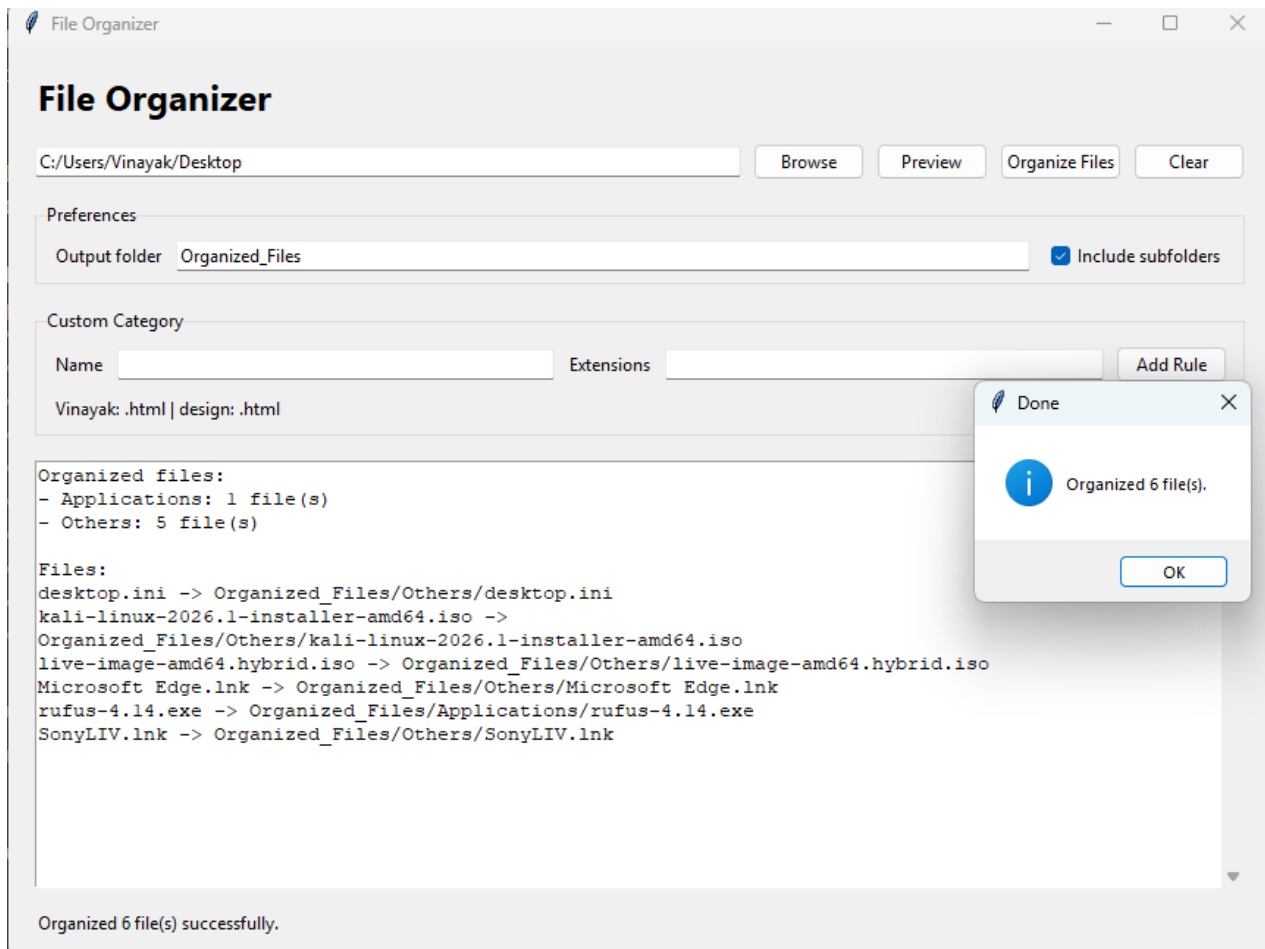


Figure 3 : GUI

6 Performance Test

The Automatic File Organizer was tested to evaluate its performance, reliability, and efficiency under different scenarios. The primary objective of testing was to ensure that the application correctly categorized files, created folders automatically, and moved files without data loss while maintaining low memory usage and fast execution. The project was also tested to verify its ability to handle different file types and empty directories.

6.1 Test Plan/ Test Cases

The following test cases were executed during project testing:

Test Case	Expected Result	Status
Organize mixed file types	Files moved to correct folders	Passed
Create missing folders	Required folders created automatically	Passed
Empty directory	Program completes without errors	Passed
Unsupported file type	File moved to "Others" folder	Passed
Large number of files	Files organized successfully	Passed
Duplicate folder names	Existing folders reused correctly	Passed

6.2 Test Procedure

- Create a test directory containing different file types.
- Run the Python application.
- Scan all files in the selected folder.
- Identify file extensions.
- Create destination folders if they do not exist.
- Move files into their respective folders.
- Verify that all files are correctly organized.
- Check for errors or missing files after execution.

6.3 Performance Outcome

The application successfully organized files into their respective folders with 100% accuracy during testing. It required minimal system resources and completed file organization quickly for normal-sized directories. No data loss or file corruption was observed during testing. The modular implementation using Python's standard libraries ensured reliable performance and easy maintenance. Overall, the application met all functional requirements and demonstrated efficient performance for real-world file organization tasks.

7 My learnings

This internship provided me with valuable practical experience in software development and significantly enhanced my technical and professional skills. During the development of the **Automatic File Organizer using Python**, I gained hands-on experience in Python programming, file handling, modular programming, debugging, testing and project documentation. I also learned how to organize a software project using multiple modules, manage source code using Git and GitHub and prepare technical reports following industry standards.

Working on this project strengthened my problem-solving and logical thinking abilities. I understood the importance of writing clean, reusable, and maintainable code while following a structured software development process. The internship also improved my communication, time management, and documentation skills through regular project planning and reporting.

Overall, this internship has increased my confidence in developing real-world software applications and has prepared me for future internships, hackathons, and software engineering roles. The knowledge and practical experience gained during this internship will contribute significantly to my academic growth and future career in software development.

8 Future work scope

The current version of the **Automatic File Organizer using Python** successfully organizes files based on their extensions. However, several enhancements can be implemented in the future to improve its functionality and user experience.

The project can also be extended to monitor folders in real time and automatically organize newly added files without requiring manual execution. Artificial Intelligence and Machine Learning techniques can be integrated to classify files based on their content instead of relying only on file extensions. Cloud storage support can be added to organize files stored on platforms such as Google Drive and OneDrive. Additional features such as duplicate file detection, automatic backup before file movement, file encryption, advanced search and filtering, customizable file categories, and activity logs can further improve the application's usability and reliability.

These enhancements would make the application more intelligent, scalable, and suitable for both personal and professional file management environments.